Generated report 1: pbmc

Objective (as given)

"Learn a low-dimensional representation and generate realistic synthetic cells for sharing"

Agent's decision

Task: representation_learning | decision: proceed | task run: representation_learning | also: data_generation

profile.json reports 2,700 cells (well above the 200-sample floor) and raw non-negative integer UMI counts (is_nonneg_integer: true, 97.4% sparsity) over 32,738 genes. Neither task needs labels, so has_labels: false is not a blocker. A latent-variable count model (e.g. a negative-binomial VAE) delivers both the low-dimensional representation and a generator in one model, which a simpler method like PCA cannot do for realistic count generation; PCA is still a useful reference for the representation. With only 2,700 cells, DATA_DESCRIPTION.md warns about memorization, so evaluation must compare against held-out real cells and include nearest-neighbor distance checks.

Models run (headline metrics; details on the metrics page)

Run names are <model id>_s<seed>: vae_nb_s0 is model vae_nb trained with seed 0. Models in this report: vae_nb (seeds 0, 1); flow_pca (seeds 0, 1). The next page explains seeds, the evaluation space, references, and memorization.

```
vae_nb_s0          vae    C2ST 0.9259 (real-vs-real 0.5855)  precision 0.7907  recall 0.8074  DCR ratio 1.4436
vae_nb_s1          vae    C2ST 0.9045 (real-vs-real 0.5664)  precision 0.8741  recall 0.8  DCR ratio 1.4175
flow_pca_s0        flow   C2ST 0.8635 (real-vs-real 0.5855)  precision 0.5593  recall 0.9833  DCR ratio 1.6254
flow_pca_s1        flow   C2ST 0.8349 (real-vs-real 0.5664)  precision 0.5444  recall 0.9815  DCR ratio 1.5725
```

Downstream-use verdicts

- [conditional] Use the NB-VAE 10-d latent codes as the input to clustering or neighbor graphs of these cells (vae_nb_s0)

- [not_recommended] Use NB-VAE synthetic cells as a stand-in for real cells in joint-distribution analyses (clustering, cell-type proportions, gene-gene structure) (vae_nb_s0)

- [conditional] Use NB-VAE synthetic counts for marginal / count-level summaries only (per-gene means, detection rates, library-size distribution), e.g. for testing pipelines or teaching (vae_nb_s0)

- [not_recommended] Use the PCA-space flow's synthetic cells as a stand-in for real cells (flow_pca_s0)

- [not_recommended] Share or release the synthetic cells externally (vae_nb_s0)

Summary (agent-written)

The objective asked for a low-dimensional representation of the PBMC cells and realistic synthetic cells for sharing. The agent trained a negative-binomial VAE on raw counts (representation and generator) and a RealNVP flow on 20 PCA components (generator comparator), each with two seeds. Neither half of the success criterion was met. The VAE latent preserves held-out neighborhoods slightly worse than a same-size PCA (trustworthiness

Generated report 1: pbmc (cont.)

0.8033 vs PCA baseline 0.8328 for vae_nb_s0). A classifier separates synthetic from held-out real cells easily for every model (C2ST AUC 0.9259 for vae_nb_s0 vs real-vs-real reference 0.5855; 0.8635 for flow_pca_s0 vs the same reference). No near-copies of training cells were found, but there is no formal privacy guarantee. External sharing is not recommended, and the VAE's synthetic counts are only conditionally usable for marginal, count-level summaries.

How to read this report

Run names and seeds

Run names are <model id>_s<seed>: vae_nb_s0 is model vae_nb trained with seed 0. Models in this report: vae_nb (seeds 0, 1); flow_pca (seeds 0, 1).

A seed fixes the model's random initialization, mini-batch order, the validation slice used for early stopping, and the random draws when sampling; the held-out split and preprocessing are the same for every run. Models with two seeds are trained twice as a stability check: a conclusion that holds for both seeds does not depend on one lucky or unlucky initialization. Two runs are far too few to estimate run-to-run variability, so the spread between seeds is not a confidence interval and should not be read as one.

Where realism is measured: the evaluation space

Synthetic and held-out real samples are compared in one shared space: the first 14 principal components (PCs) of the transformed data, fit on the training split. These 14 PCs explain 22.6% of the variance of the transformed training data, so the realism metrics (C2ST, MMD, precision/recall, DCR) say nothing about the remaining 77.4%. Per-feature statistics are computed on all features instead.

Why only 14: PCA fitted to the training data overfits it in later components, which would let a classifier tell training data from new real data apart and bias every comparison. The pipeline keeps the components that hold up on unseen data (rule: largest k where an unseen half of the training data keeps >= 0.85 of the fitted half's cumulative PCA variance).

Reference values

Every realism metric is shown with a real-vs-real reference: the same metric computed between a subsample of the training data and the held-out real data. That is the value two genuinely real samples of this size give, so a model is judged by how close it gets to the reference, not to a theoretical ideal.

How memorization is characterized

For each sample, DCR (distance to closest record) is the distance to its nearest training sample in the evaluation space. The held-out real samples give the yardstick: how close a new sample from the same population typically lands to some training sample. dcr_ratio compares the median synthetic DCR to the median held-out DCR (about 1 = as far as new real data; well below 1 = suspiciously close). too_close_fraction counts synthetic samples closer than the 5th percentile of held-out DCRs (about 0.05 is expected without copying).

This detects near-copies of training samples. It is not a privacy guarantee: it does not test membership inference, attribute inference, or any formal property such as differential privacy.

How the learned representation is judged

Trustworthiness (0 to 1) asks whether samples that are neighbors in the latent space are also neighbors in the full transformed data; it is penalized when the latent space puts unrelated samples together. It is compared with a PCA of the same size, fit on the training data, which is the simplest alternative representation.

How to read this report (cont.)

Caution when comparing flow_pca with the other models

flow_pca is trained on the top 20 PCA components of the transformed data, and realism is scored in a 14-component PCA of the same transformed data. The directions it models directly are largely the directions the realism metrics look at, while vae_nb must reproduce them through a model of all 2000 features. The shared realism metrics may therefore favor it. Per-feature and count-level checks, which use all features, are a fairer comparison of what each model reproduces outside those components.

Objective interpretation

Interpretation

Primary task: representation_learning

Secondary tasks: data_generation

'Learn a low-dimensional representation' maps to representation_learning, which is listed first and is the primary task; 'and generate realistic synthetic cells' explicitly asks for a second thing, so data_generation is a secondary task. 'for sharing' qualifies the generation goal (release of synthetic cells), which raises memorization checks to a required part of the evaluation.

Success criterion

Representation: held-out reconstruction likelihood is close to the training value and the latent space preserves nearest-neighbor structure (trustworthiness) of the log-normalized held-out cells. Generation: a classifier cannot tell synthetic from held-out real cells well (C2ST AUC close to its real-vs-real reference), and synthetic cells are no closer to training cells than held-out real cells are.

Whether to model, and how

Decision: proceed (task actually run: representation_learning)

profile.json reports 2,700 cells (well above the 200-sample floor) and raw non-negative integer UMI counts (is_nonneg_integer: true, 97.4% sparsity) over 32,738 genes. Neither task needs labels, so has_labels: false is not a blocker. A latent-variable count model (e.g. a negative-binomial VAE) delivers both the low-dimensional representation and a generator in one model, which a simpler method like PCA cannot do for realistic count generation; PCA is still a useful reference for the representation. With only 2,700 cells, DATA_DESCRIPTION.md warns about memorization, so evaluation must compare against held-out real cells and include nearest-neighbor distance checks.

Other readings considered

- The objective could be read with data_generation as primary (because of 'for sharing'); representation_learning was chosen as primary because it is named first, and generation is kept as an explicit secondary task so both are evaluated.

- 'For sharing' is privacy-relevant wording: this pipeline only checks memorization via nearest-neighbor distances to training cells, which is evidence, not a formal privacy guarantee such as differential privacy. No claim will be made that the synthetic cells are private, anonymized, or safe to share.

- 'Realistic' is not defined by the user; it is operationalized here as C2ST AUC against held-out real cells relative to a real-vs-real reference, plus per-gene marginal statistics, on the preprocessed gene subset rather than all 32,738 genes.

Data and preprocessing

Dataset (from DATA_DESCRIPTION.md)

Peripheral blood mononuclear cells (PBMCs) from a healthy donor, profiled with 10x Genomics single-cell RNA sequencing (scRNA-seq). This is the "pbmc3k" dataset used throughout the Scanpy tutorials and widely used as a benchmark for single-cell analysis methods. Stored as the raw h5ad file from the Scanpy tutorials.

Profile

2,700 samples x 32,738 features; modality tabular

Sparsity 0.9741; value range [0, 419]; non-negative integers: True

Labels: none

Held-out split

test_fraction 0.2, stratified: False -> 2160 training / 540 held-out samples

Reason: With 2,700 cells, 0.2 holds out 540 real cells, enough for C2ST and nearest-neighbor memorization checks against held-out data, while keeping 2,160 cells for training (the description warns that with so few cells large models can memorize, so a sizeable held-out set matters). stratify is false because profile.json reports has_labels: false.

Preprocessing (applied identically for every model; fit on training data only)

- select_top_variance {"n_features": 2000}: profile.json reports 32,738 genes at 97.4% sparsity; most genes are near-all-zero in 2,700 cells and add noise and training time. Keeping the 2,000 most variable genes (ranked on normalized, log-transformed training cells only) keeps the informative signal. Judgment call: 2,000 vs 1,000-5,000; 2,000 matches the timing table and is the common scale for pbmc3k-size data.

- normalize_total : DATA_DESCRIPTION.md says library size varies about 10x between cells (mean ~2,367 counts); normalizing per cell removes sequencing depth from the transformed view so that cell-to-cell distances in the evaluation space reflect expression, not depth.

- log1p : The counts are heavy-tailed (max 419, mean 0.072 per entry); log1p compresses the few highly expressed genes that would otherwise dominate the PCA evaluation space and the flow's PCA input. The NB-VAE still trains on raw counts.

Evaluation space

All models are scored in one shared space: PCA of the transformed data with 14 components (0.2258 of training variance), fit on the training split. The number of components is the largest k where an unseen half of the training data keeps >= 0.85 of the fitted half's cumulative PCA variance: a PCA with more components overfits the training data, and a classifier could then tell training from held-out real samples apart, biasing every comparison.

Models, hyperparameters, and training

vae_nb

Family: vae; input: raw; seeds [0, 1]; serves representation_learning, data_generation; status planned; estimated 1.5 min/seed

Why this model: The objective asks for both a low-dimensional representation and synthetic cells; an NB-VAE on raw counts gives a 10-d latent embedding and a count generator in one model, matching the raw UMI data in profile.json. Its role does not depend on beating PCA: revision 1 showed its latent is slightly less trustworthy than a same-size PCA (0.80 vs 0.83), so the report compares it to that PCA baseline and the VAE's distinct contribution is that it generates count data (library sizes, zero fractions) that PCA cannot. Unchanged in revision 2. Two seeds because the sharing claim depends on whether fidelity and memorization results are stable, and the budget allows it. Compared against the PCA flow in the common evaluation space.

- latent_dim = 10: n_train is about 2,160 cells (< 5,000); a 10-dimensional latent space is enough for the handful of major PBMC populations and is easy to inspect, while larger latent sizes would mostly fit noise in this few cells.

- hidden_dims = [256, 128]: The input is 2,000 selected genes (>= 1,000 features), so [256, 128] compresses 2,000 -> 10 in two steps without a huge parameter count relative to 2,160 training cells.

- epochs = 400: Maximum only; early stopping on the 10% validation split (patience 25) ends training sooner. 400 is affordable: the timing table puts ~2,000 cells x 2,000 genes at 0.9 min for 400 epochs.

- batch_size = 128: With ~1,944 cells after the validation carve-out, 128 gives about 15 steps per epoch, enough gradient updates per epoch on this small dataset.

- lr = 0.001: Adam at 1e-3 has no prior loss curve on this run suggesting oscillation; it will be lowered only if the recorded curve oscillates.

- likelihood = nb: profile.json reports is_nonneg_integer: true with 97.4% zeros and DATA_DESCRIPTION.md recommends count likelihoods; a library-size-aware negative binomial decoder models raw UMI overdispersion without the distortion of log-transforming sparse counts.

- kl_warmup_epochs = 40: About 10% of the 400-epoch maximum; ramping KL from 0 avoids posterior collapse early, which matters because the representation is the primary deliverable.

Training (vae_nb_s0): 99 of 400 max epochs, kept epoch 74 (validation loss did not improve for 25 epochs); 0.192 min on cpu; 1944 fit / 216 validation samples; 1,099,700 parameters

Training (vae_nb_s1): 105 of 400 max epochs, kept epoch 80 (validation loss did not improve for 25 epochs); 0.202 min on cpu; 1944 fit / 216 validation samples; 1,099,700 parameters

flow_pca

Family: flow; input: pca; seeds [0, 1]; serves data_generation; status planned; estimated 0.5 min/seed

Models, hyperparameters, and training (cont.)

Why this model: Comparator for the generation half of the objective: an exact-likelihood model with a different inductive bias (continuous density on log-normalized PCA rather than a count decoder), scored in the same PCA evaluation space, so the report can tell whether the VAE's synthetic cells are good relative to an alternative. It provides no latent representation, so it serves data_generation only. Two seeds for the same stability reason as the VAE.

- pca_components = 20: With n_train about 2,160 cells, density estimation needs many cells per dimension, so 20 components (upper end of the 10-20 range). Revision 1 showed these 20 components explain only 24% of the transformed variance (input_pca_variance_explained 0.2396), so the flow models a small share of the signal and treats the remaining 76% as independent Gaussian noise; that is a limitation. Judgment call: kept at 20 rather than raised, because revision 1 already overfit by epoch 13-14 at 20 dimensions and more dimensions per cell would make that worse.

- n_layers = 6: Reduced from 8 to 6 coupling layers: in revision 1 the 8-layer flow (194,880 parameters for 1,944 fit cells) reached its best validation NLL at epoch 13-14 and then overfit sharply. 6 layers is the lower end of the usual range and still captures a multi-modal 20-d density.

- hidden_dim = 64: Reduced from 128 to 64 for the same observed overfitting (val NLL 27.3 -> 39.1 while train NLL kept falling in flow_pca_s0); a smaller coupling network fits 1,944 cells in 20 dimensions with less capacity to memorize.

- epochs = 400: Maximum only with early stopping; with the lower learning rate, more epochs are needed to converge, and 400 still costs well under a minute at this size.

- batch_size = 128: 128 gives about 15 steps per epoch on ~1,944 training cells after the validation carve-out.

- lr = 0.0003: Lowered from 1e-3 to 3e-4 because revision 1's loss curves show validation NLL bottoming out at epoch 13-14 and then rising steeply, i.e. too-fast fitting on this small 20-d PCA input.

Training (flow_pca_s0): 87 of 400 max epochs, kept epoch 62 (validation loss did not improve for 25 epochs); 0.044 min on cpu; 1944 fit / 216 validation samples; 48,624 parameters; input PCA keeps 0.2396 of variance

Training (flow_pca_s1): 84 of 400 max epochs, kept epoch 59 (validation loss did not improve for 25 epochs); 0.043 min on cpu; 1944 fit / 216 validation samples; 48,624 parameters; input PCA keeps 0.2396 of variance

Evaluation focus

Representation: held-out vs training reconstruction loss (generalization gap) and trustworthiness of the VAE latent space with respect to log-normalized held-out cells. Generation: C2ST AUC of synthetic vs held-out real cells relative to its real-vs-real reference, marginal per-gene statistics, and nearest-neighbor distance ratios (synthetic-to-train vs held-out-to-train) as memorization evidence, which matters most because the objective mentions sharing; memorization checks are evidence, not a privacy guarantee. Raw data is unfiltered (no QC), which is a limitation.

Evaluation metrics (held-out real data)

Realism metrics use the 14-component evaluation space (22.6% of transformed-data variance); per-feature statistics use all features. Each realism metric has a real-vs-real reference row.

Fidelity

	vae_nb_s0	vae_nb_s1	flow_pca_s0	flow_pca_s1
n_compared	540	540	540	540
mmd2	0.01703	0.01528	0.00772	0.00589
mmd2_reference	0.00026	-0.00028	0.00026	-0.00028
c2st_auc	0.9259	0.9045	0.8635	0.8349
c2st_auc_reference	0.5855	0.5664	0.5855	0.5664
precision	0.7907	0.8741	0.5593	0.5444
recall	0.8074	0.8	0.9833	0.9815

Memorization

	vae_nb_s0	vae_nb_s1	flow_pca_s0	flow_pca_s1
dcr_median_synthetic	2.9292	2.8762	3.2981	3.1906
dcr_median_heldout	2.029	2.029	2.029	2.029
dcr_ratio	1.4436	1.4175	1.6254	1.5725
too_close_fraction	0	0.0019	0.0019	0

Evidence about copying of training samples, not a formal privacy guarantee.

Feature stats

	vae_nb_s0	vae_nb_s1	flow_pca_s0	flow_pca_s1
mean_corr	0.9978	0.9977	0.9983	0.9984
var_corr	0.8928	0.9007	0.9709	0.9794
zero_frac_corr	0.9919	0.9917	n/a	n/a
zero_frac_synthetic	0.7359	0.7379	n/a	n/a
zero_frac_heldout	0.7232	0.7232	n/a	n/a
library_median_ratio	0.9971	0.987	n/a	n/a

Likelihood

	vae_nb_s0	vae_nb_s1	flow_pca_s0	flow_pca_s1
train_mean	-1556.868	-1551.39	-25.845	-25.836
heldout_mean	-1586.624	-1588.151	-24.569	-24.631
per_dimension_heldout	-0.7933	-0.7941	-1.2284	-1.2315

Likelihoods are in each model's own input space and units; do not compare them across models with different input spaces.

Representation

	vae_nb_s0	vae_nb_s1	flow_pca_s0	flow_pca_s1
latent_dim	10	10	n/a	n/a
trustworthiness	0.8033	0.7941	n/a	n/a
trustworthiness_pca_baseline	0.8328	0.8327	n/a	n/a
knn_overlap	0.1607	0.1567	n/a	n/a

Evaluation metrics (held-out real data) (cont.)

knn_overlap_pca_baseline	0.1681	0.1677	n/a	n/a

Neighborhoods are Euclidean in the transformed space, which favors PCA (it is the best linear Euclidean summary by construction); a latent close to the baseline is not a failure.

How to read the metrics

- mmd2: Squared maximum mean discrepancy (RBF kernel, bandwidth = median distance among held-out real samples) between synthetic and held-out real samples in the evaluation PCA space. 0 = same distribution; compare with mmd2_reference. The estimator is unbiased, so values near 0 can be slightly negative.

- mmd2_reference: Same MMD, but a training-set subsample vs held-out real: the value two real samples of this size give.

- c2st_auc: Classifier two-sample test: cross-validated ROC AUC of a gradient-boosted classifier separating synthetic from held-out real samples. 0.5 = indistinguishable, 1.0 = perfectly separable.

- c2st_auc_reference: Same test, training subsample vs held-out real (should be near 0.5).

- precision: Fraction of synthetic samples inside the 5-NN radius of some held-out real sample (fidelity: are samples realistic?).

- recall: Fraction of held-out real samples inside the 5-NN radius of some synthetic sample (coverage: is the data's variety reproduced?).

- dcr_median_synthetic: DCR = distance to closest record: for one sample, the Euclidean distance (evaluation space) to its nearest training sample. This is the median DCR over synthetic samples.

- dcr_median_heldout: Median DCR over held-out real samples: how close a genuinely new sample from the same population typically is to some training sample. This is the yardstick for the synthetic DCR.

- dcr_ratio: dcr_median_synthetic / dcr_median_heldout. About 1 means synthetic samples are as far from the training data as new real samples are; well below 1 means they sit unusually close to training samples (possible copying); above 1 means they are farther away than real data (often a sign of off-distribution samples, not of privacy).

- too_close_fraction: Fraction of synthetic samples whose DCR is below the 5th percentile of held-out DCRs. If the model does not copy, about 0.05 is expected (the same share as for new real samples); much higher means a group of synthetic samples are near-copies of training samples. Evidence about copying only, not a privacy guarantee.

- mean_corr: Pearson correlation, across features, between per-feature means of synthetic and held-out real data (transformed space).

- var_corr: Same, for per-feature variances.

- zero_frac_corr: Same, for each feature's fraction of exact zeros (raw counts; only for models that generate counts).

- library_median_ratio: Median total count per synthetic sample divided by the held-out median (counts only).

- train_mean: Mean per-sample ELBO (VAE, a lower bound) or exact log-likelihood (flow) on training data, in the model's own input space. Not comparable across models with different input spaces.

How to read the metrics (cont.)

- heldout_mean: Same quantity on held-out real data; a large drop from train_mean indicates overfitting.

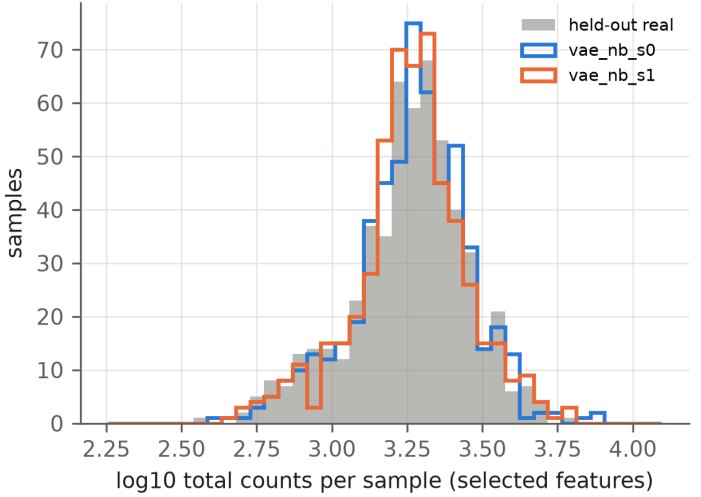
- trustworthiness: Trustworthiness (Venna and Kaski, 2001), between 0 and 1: for each held-out sample, look at its 15 nearest neighbors in the latent space and penalize any that are NOT among its 15 nearest neighbors in the reference space (full transformed data), more heavily the farther away they really are. 1 = every latent-space neighbor is a true neighbor; lower = the latent space puts unrelated samples next to each other.

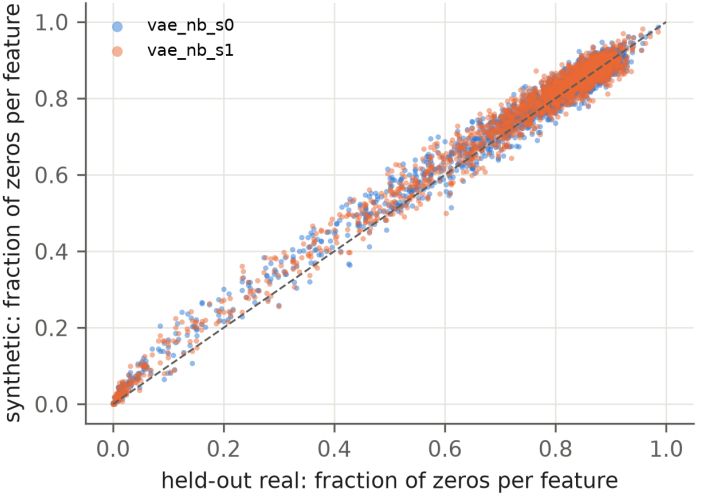
- trustworthiness_pca_baseline: Trustworthiness of a PCA (fit on training data) with as many dimensions as the latent space: the bar a learned representation is compared with.

- knn_overlap: Average fraction of each held-out sample's 15 nearest neighbors in the full transformed space that are also among its 15 nearest neighbors in the latent space.

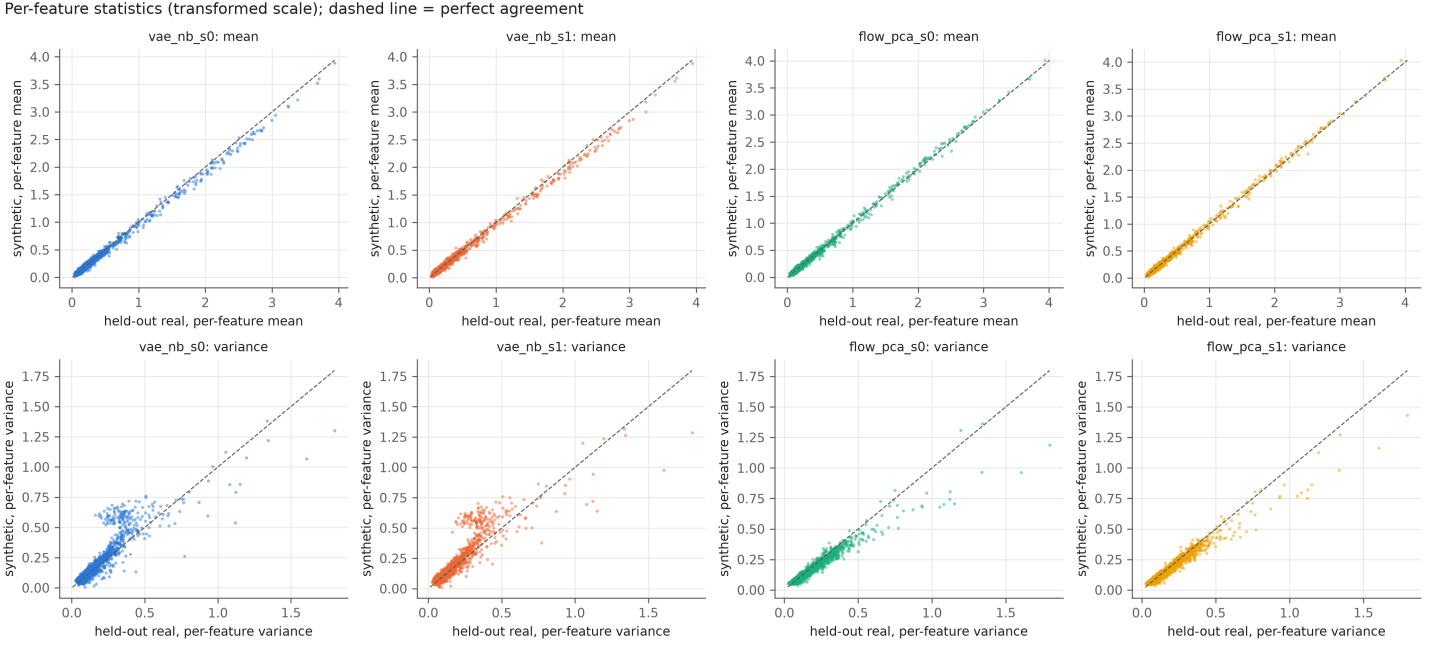
- knn_overlap_pca_baseline: Same, for the same-size PCA baseline.

Count-level properties

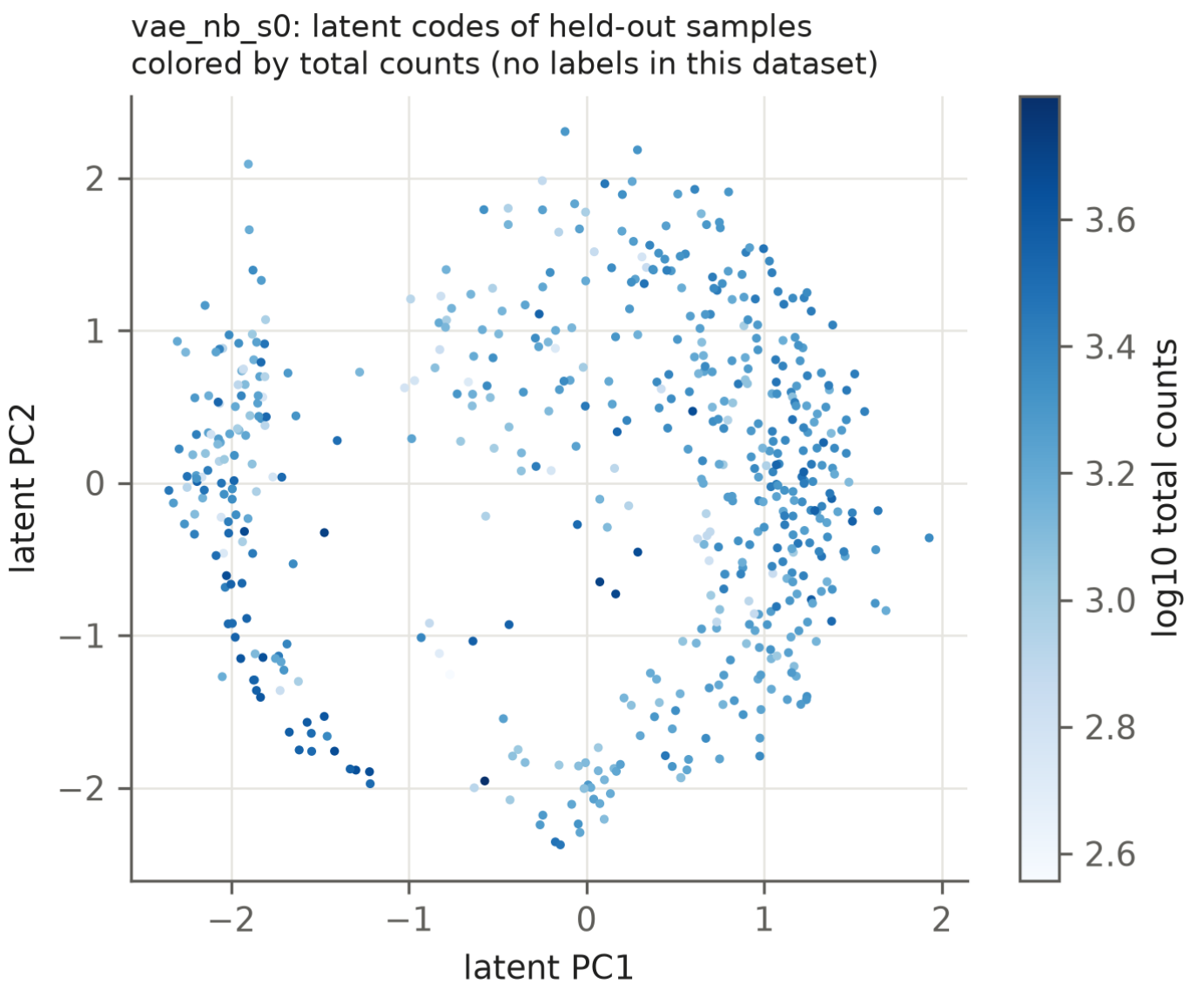




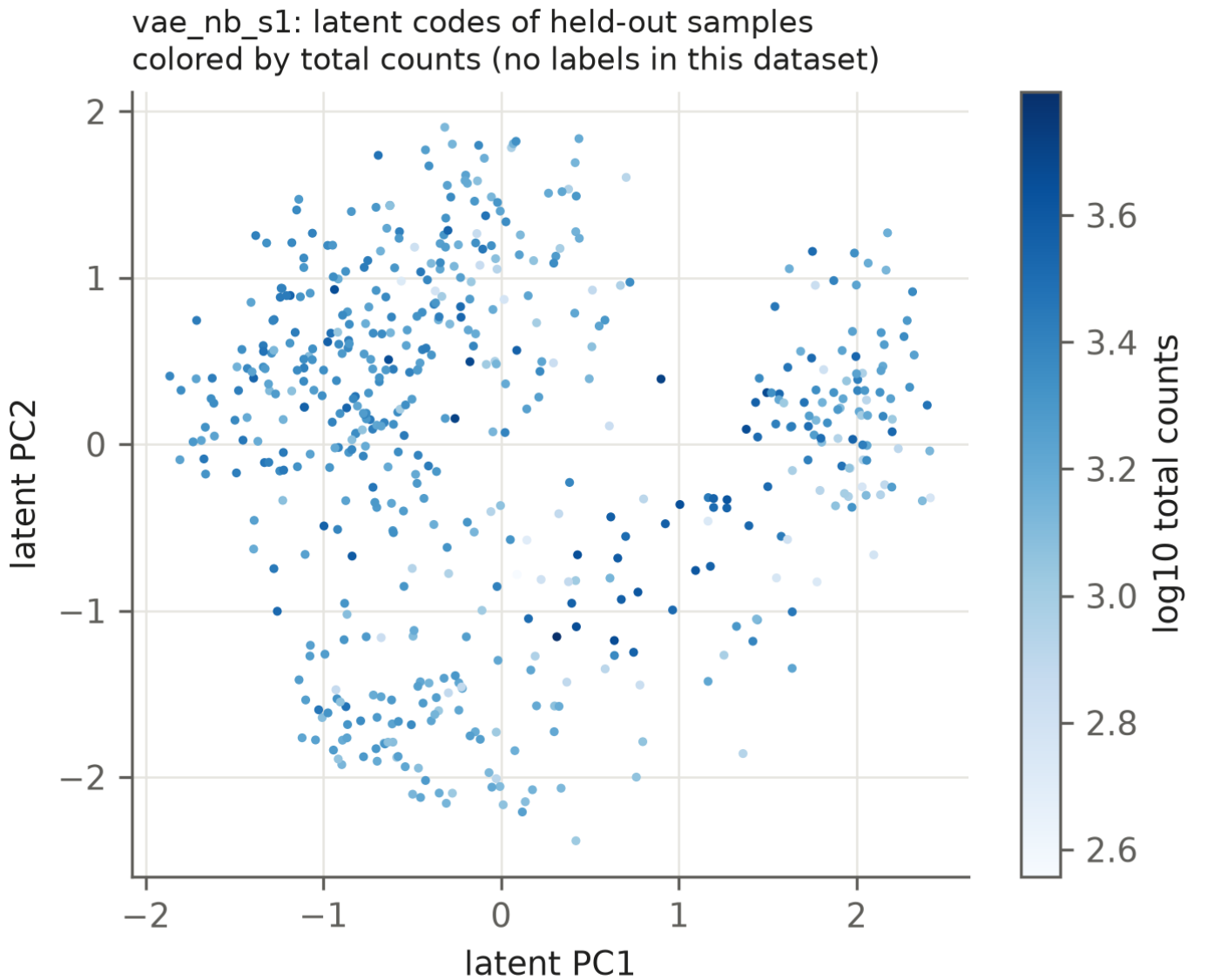
Count-level checks for models that generate counts: total counts per sample, and per-feature fraction of zeros.



Per-feature mean and variance of synthetic vs held-out real data on the transformed scale. Points on the dashed line agree exactly.

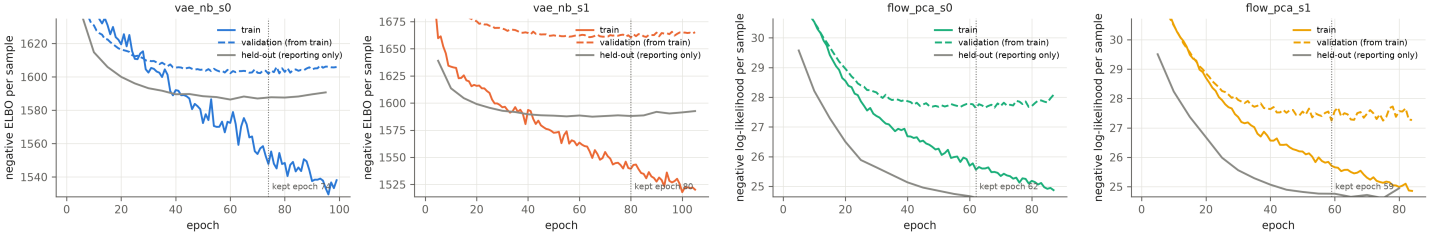


Two-dimensional PCA of a VAE's latent codes for held-out samples. A picture for orientation; neighbor-preservation numbers are on the metrics page.

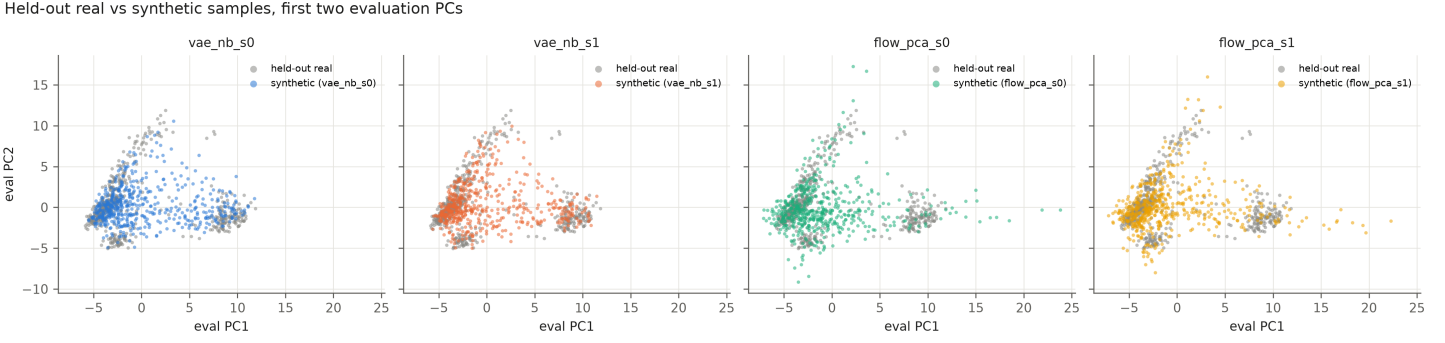


Two-dimensional PCA of a VAE's latent codes for held-out samples. A picture for orientation; neighbor-preservation numbers are on the metrics page.

Training curves



Training curves. The validation set is 10% of the training split and decides early stopping; the held-out curve is shown for reference only and never used to choose anything.



Held-out real samples (gray) and synthetic samples in the first two principal components of the shared evaluation space (for images: of the evaluation classifier's features). Overlap in two components is necessary but not sufficient; the C2ST and MMD metrics use all components.

Findings

What each model produced

vae_nb_s0 (NB-VAE, raw counts, 10-d latent). C2ST AUC 0.9259 (reference 0.5855); MMD2 0.01703 (reference 0.00026); precision 0.7907, recall 0.8074. Memorization: dcr_ratio 1.4436, too_close_fraction 0.0. Per-gene means correlate at 0.9978, zero fractions at 0.9919, variances at 0.8928; median library size ratio 0.9971. Held-out ELBO -1586.624 vs training -1556.868, so the model generalizes. Latent trustworthiness 0.8033 vs PCA baseline 0.8328; kNN overlap 0.1607 vs 0.1681. Training stopped early at epoch 99, keeping epoch 74.

vae_nb_s1 (second seed). C2ST AUC 0.9045 (reference 0.5664); MMD2 0.01528 (reference -0.00028); precision 0.8741, recall 0.8. dcr_ratio 1.4175, too_close_fraction 0.0019. Trustworthiness 0.7941 vs PCA baseline 0.8327. Kept epoch 80. Every conclusion from seed 0 holds on this seed.

flow_pca_s0 (RealNVP on 20 PCs, revision 2). C2ST AUC 0.8635 (reference 0.5855); MMD2 0.00772 (reference 0.00026); precision 0.5593, recall 0.9833. dcr_ratio 1.6254, too_close_fraction 0.0019. Variance correlation 0.9709. The 20 input components explain 0.2396 of the transformed variance. Kept epoch 62 of 87 run.

flow_pca_s1 (second seed, revision 2). C2ST AUC 0.8349 (reference 0.5664); MMD2 0.00589 (reference -0.00028); precision 0.5444, recall 0.9815. dcr_ratio 1.5725, too_close_fraction 0.0. Kept epoch 59 of 84 run.

In revision 1 the flow (8 layers, hidden 128, lr 1e-3) peaked at epoch 13 or 14 and then overfit. The revision-2 settings (6 layers, hidden 64, lr 3e-4) gave a stable validation curve but did not materially change fidelity: the revision-1 C2ST values recorded in the critique are close to the revision-2 values above.

What the figures show

Real vs synthetic (first two evaluation PCs): held-out real cells form a few compact groups. VAE samples cover the main groups but also fill the empty space between them, which is the smearing a classifier picks up. Flow samples spread even more widely, with a tail of points far outside the real range along PC1 and below the lower group.

Count properties (VAE only): synthetic library sizes match the held-out distribution closely, and per-gene zero fractions lie on the diagonal, with synthetic slightly more zeros for moderately detected genes.

Per-gene statistics: means agree closely for all four runs. For the VAE, a block of genes with real variance around 0.2 to 0.5 has synthetic variance near 0.6, so those genes are too noisy in the synthetic cells. The flow's variances lie closer to the diagonal, though the highest-variance genes are underestimated.

Latent space (VAE, held-out cells, colored by total counts): both seeds show several separated groups of cells, and the coloring by library size shows no strong gradient, so the latent is not dominated by sequencing depth. The dataset has no labels, so the groups cannot be named here; the structure is exploratory only.

Training curves: VAE validation loss flattens after the KL warm-up and the kept epochs sit on that plateau. The revision-2 flow curves no longer rise sharply after the best epoch.

Findings (cont.)

How the models compare

On fidelity metrics computed in the same 14-component evaluation space, the flow has lower C2ST AUC and MMD than the VAE on both seeds, with higher recall but much lower precision: it covers the real data but places many samples outside real-cell neighborhoods. The evaluation space is a PCA of the same log-normalized data the flow is trained on, which may favor the flow on these realism metrics. The VAE produces counts and matches count-level properties (zero fractions, library size), which the flow cannot produce. Neither model is better for every purpose: the VAE is the better source of count-like data, the flow covers the real data better in PCA space, and neither is realistic by the C2ST criterion. Likelihoods are not comparable across the two models, because the VAE's ELBO is on raw counts and the flow's log-likelihood is on PCA scores.

Answer to the objective

No. Representation: the VAE latent generalizes but does not beat a same-size PCA on trustworthiness (0.8033 vs 0.8328 for vae_nb_s0, 0.7941 vs 0.8327 for vae_nb_s1), so for neighbor-based uses a PCA does at least as well. Generation: every model's C2ST AUC is far above its real-vs-real reference, so the synthetic cells are not realistic as a whole. The memorization check found no copying (too_close_fraction 0.0 for vae_nb_s0, against about 0.05 expected for new real cells), but dcr_ratio above 1 means the samples sit off-distribution. This is not a privacy property, and no formal privacy guarantee exists for these cells.

Downstream use of the generated data

Use the NB-VAE 10-d latent codes as the input to clustering or neighbor graphs of these c

Verdict: CONDITIONAL | task: representation_learning | model: vae_nb_s0

The latent generalizes (held-out ELBO about 2% below training), but on both seeds it preserves held-out neighborhoods slightly worse than a 10-component PCA of the log-normalized data. For neighbor-based uses, a PCA does at least as well; the VAE latent is worth using only when a count-aware latent tied to a generator is needed.

- evidence: vae_nb_s0.representation.trustworthiness = 0.8033 (vs vae_nb_s0.representation.trustworthiness_pca_baseline = 0.8328)

- evidence: vae_nb_s1.representation.trustworthiness = 0.7941 (vs vae_nb_s1.representation.trustworthiness_pca_baseline = 0.8327)

- evidence: vae_nb_s0.representation.knn_overlap = 0.1607 (vs vae_nb_s0.representation.knn_overlap_pca_baseline = 0.1681)

- evidence: vae_nb_s0.likelihood.heldout_mean = -1586.624 (vs vae_nb_s0.likelihood.train_mean = -1556.868)

- caveat: Use only if a count-likelihood (negative binomial) latent is specifically wanted; otherwise prefer a same-size PCA, which scored higher on trustworthiness on both seeds.

- caveat: Neighborhood metrics are Euclidean in the log-normalized space, which favors PCA by construction.

- caveat: Cells were not QC-filtered, so low-quality cells and doublets are embedded too.

Use NB-VAE synthetic cells as a stand-in for real cells in joint-distribution analyses (cluste

Verdict: NOT_RECOMMENDED | task: data_generation | model: vae_nb_s0

A classifier separates VAE cells from held-out real cells with AUC above 0.9 against a real-vs-real reference below 0.59, and the real-vs-synthetic PCA plot shows synthetic cells filling the gaps between real populations. The joint distribution is not reproduced well enough to stand in for real cells.

- evidence: vae_nb_s0.fidelity.c2st_auc = 0.9259 (vs vae_nb_s0.fidelity.c2st_auc_reference = 0.5855)

- evidence: vae_nb_s1.fidelity.c2st_auc = 0.9045 (vs vae_nb_s1.fidelity.c2st_auc_reference = 0.5664)

- evidence: vae_nb_s0.fidelity.mmd2 = 0.01703 (vs vae_nb_s0.fidelity.mmd2_reference = 0.00026)

- caveat: Both seeds agree; the gap to the reference is far larger than seed-to-seed variation.

Use NB-VAE synthetic counts for marginal / count-level summaries only (per-gene means,

Verdict: CONDITIONAL | task: data_generation | model: vae_nb_s0

Per-gene means, zero fractions and library sizes match held-out real cells closely, so the

Downstream use of the generated data (cont.)

VAE output is usable where only marginal count properties matter, e.g. exercising a scRNA-seq pipeline. Per-gene variances are less accurate: a group of genes has inflated variance in the feature-statistics figure.

- evidence: vae_nb_s0.feature_stats.mean_corr = 0.9978 (vs 1.0 = identical per-gene means to held-out real cells)
- evidence: vae_nb_s0.feature_stats.zero_frac_corr = 0.9919 (vs vae_nb_s0.feature_stats.zero_frac_heldout = 0.7232 vs zero_frac_synthetic = 0.7359)
- evidence: vae_nb_s0.feature_stats.library_median_ratio = 0.9971 (vs 1.0 = same median library size as held-out real cells)
- evidence: vae_nb_s0.feature_stats.var_corr = 0.8928 (vs flow_pca_s0.feature_stats.var_corr = 0.9709)
- caveat: Only for uses that do not depend on cell-cell or gene-gene joint structure (C2ST AUC above 0.9).
- caveat: Covers the 2,000 selected genes only, not all 32,738.
- caveat: Not for external release without a privacy review; see privacy_statement.

Use the PCA-space flow's synthetic cells as a stand-in for real cells

Verdict: NOT_RECOMMENDED | task: data_generation | model: flow_pca_s0

The flow is somewhat harder to separate from real cells than the VAE, but its C2ST AUC is still far above the reference, and only about 55% of its samples fall within real-cell neighborhoods (it covers the data but scatters many samples off-manifold). Its samples are continuous log-normalized values, not counts, and the evaluation space is close to its own input space, which favors it.

- evidence: flow_pca_s0.fidelity.c2st_auc = 0.8635 (vs flow_pca_s0.fidelity.c2st_auc_reference = 0.5855)
- evidence: flow_pca_s1.fidelity.c2st_auc = 0.8349 (vs flow_pca_s1.fidelity.c2st_auc_reference = 0.5664)
- evidence: flow_pca_s0.fidelity.precision = 0.5593 (vs vae_nb_s0.fidelity.precision = 0.7907)
- evidence: flow_pca_s0.fidelity.recall = 0.9833 (vs vae_nb_s0.fidelity.recall = 0.8074)
- caveat: The revision-2 retrain removed the early overfitting but did not change C2ST materially.

Share or release the synthetic cells externally

Verdict: NOT_RECOMMENDED | task: data_generation | model: vae_nb_s0

No near-copies of training cells were found in the evaluation space, but the large distance to training cells reflects samples that are off-distribution, not a privacy property. The cells also fail the realism criterion, so they do not meet the 'realistic cells for sharing'

Downstream use of the generated data (cont.)

goal. Release would additionally need a formal privacy review or a formally private training method.

- evidence: vae_nb_s0.memorization.too_close_fraction = 0 (vs about 0.05 expected for new real cells)

- evidence: vae_nb_s0.memorization.dcr_ratio = 1.4436 (vs 1.0 = as far from training cells as new real cells)

- evidence: vae_nb_s0.fidelity.c2st_auc = 0.9259 (vs vae_nb_s0.fidelity.c2st_auc_reference = 0.5855)

- caveat: Memorization was checked only in a 14-component PCA space, on 540 synthetic cells per model.

- caveat: No formal privacy guarantee of any kind.

Not tested (no recommendation either way)

- Cell-type-level fidelity (does each PBMC population appear in the right proportion): the dataset has no labels; it would need annotated cells or a marker-based annotation applied identically to real and synthetic cells.

- Downstream analysis equivalence (e.g. differential expression or clustering results reproduced on synthetic cells): would need running the same analysis on real and synthetic data and comparing the outputs.

- Fidelity outside the 14 evaluation PCs (e.g. C2ST on all 2,000 genes, gene-gene correlation agreement): not computed by this pipeline.

- Membership-inference or attribute-inference attacks: not run; DCR-based memorization is the only leakage evidence.

Privacy

There is no formal privacy guarantee (such as differential privacy) for any of these synthetic cells. The only check run was memorization: distance from each synthetic cell to its nearest training cell in a 14-component PCA space. No near-copies were found (too_close_fraction 0.0 for vae_nb_s0, 0.0019 for vae_nb_s1, 0.0019 for flow_pca_s0, 0.0 for flow_pca_s1, against about 0.05 expected for new real cells), and median distances are 1.42 to 1.63 times those of held-out real cells. That shows the models did not copy training cells in this space. It says nothing about membership inference, rare-cell leakage, or leakage through genes outside the evaluated subspace, and the large distances mainly reflect that the samples are off-distribution. Any release requires a proper privacy review or a formally private training method.

Critique and revision

An independent critic subagent reviewed the objective, plan, metrics, and training logs (not the agent's own reasoning) before the recommendation and findings were written.

Neither half of the success criterion is met, and the plan's main rationale is contradicted. The NB-VAE latent is less trustworthy than a same-size PCA on both seeds (0.803/0.794 vs 0.833), and every model's C2ST AUC (0.84-0.93) is far above its real-vs-real reference (0.57-0.59), so the synthetic cells are clearly distinguishable from real ones. The flow, which looks better on C2ST and MMD, stopped improving at epoch 13-14 and then overfit sharply. It is also scored in a PCA space nearly identical to its own input space, so the flow-vs-VAE comparison favors the flow. The memorization metrics show no copying (too_close_fraction 0-0.007), but dcr_ratio of 1.42-1.59 means the samples are off-distribution, not that they are safe. No metric supports a privacy or sharing claim. All seeds, references, run logs and plan reasons were checked; seed variation is smaller than the gaps between models.

[high] objective test

The representation half of the success criterion (trustworthiness of the latent) is measured and fails: the NB-VAE latent is worse than a 10-component PCA on both seeds and on both neighborhood metrics. Any conclusion that the VAE gives a good or better low-dimensional representation is unsupported. PCA is the better representation by the plan's own metric.

Evidence: metrics/vae_nb_s0.json: trustworthiness 0.8033 vs trustworthiness_pca_baseline 0.8328, knn_overlap 0.1607 vs 0.1681. metrics/vae_nb_s1.json: 0.7941 vs 0.8327, 0.1567 vs 0.1677. The VAE's seed-to-seed difference (0.009) is much smaller than its gap to PCA (0.03-0.04). The generalization part does pass: ELBO train -1556.9 vs held-out -1586.6 (s0), -1551.4 vs -1588.2 (s1), a gap of about 2%.

Recommendation: Report PCA as the representation that best preserves neighbors. The metric note says Euclidean neighborhoods favor PCA; if the VAE is kept as the representation, justify it on a criterion it actually wins (e.g. count-aware reconstruction), or add a criterion such as a neighborhood metric in a count-aware reference space. Otherwise state that the VAE latent does not beat PCA.

[high] objective test

The generation half of the success criterion ('C2ST AUC close to its real-vs-real reference') fails for every model and seed. A classifier separates synthetic from held-out real cells with AUC 0.84-0.93, against references of 0.57-0.59. 'Realistic synthetic cells' is not established. Picking the less-bad model as the generator does not make its cells realistic.

Evidence: c2st_auc / c2st_auc_reference: vae_nb_s0 0.9259/0.5855, vae_nb_s1 0.9045/0.5664, flow_pca_s0 0.8572/0.5855, flow_pca_s1 0.8352/0.5664. MMD2 is 30-60x its reference: vae 0.01703 and 0.01528, flow 0.00767 and 0.00959, vs references 0.00026 and -0.00028.

Recommendation: State that the criterion was not met. Within budget (the runs took 0.03-0.2 min each), try generators aimed at the failure, e.g. the flow with lower lr and a smaller network (see training finding), or more PCA components. Re-evaluate before making any 'realistic' claim.

Critique and revision (cont.)

[medium] evaluation-space bias

Fidelity and memorization are computed in a 14-component PCA of the transformed training data. The flow models a 20-component PCA of the same transformed data, so the evaluation space is essentially a subspace of the flow's own input. That favors the flow over the count-space VAE in C2ST, MMD and DCR. The 14-PC space also leaves out most of the variance (the flow's 20 PCs explain only 24%), so the flow's 'residual as independent Gaussian noise' and the gene-gene structure outside the top PCs go untested.

Evidence: metrics/*.json eval_space: 'PCA (14 components) of transformed data, fit on training split'. runs/flow_pca_s0.json and s1: input_pca_variance_explained 0.2396, n_in 20. plan.json flow_pca pca_components reason: 'treats the residual variance outside them as independent Gaussian noise when generating full-dimensional samples'.

Recommendation: Caveat any flow-over-VAE ranking. Add a fidelity check outside the top PCs (e.g. C2ST on the full transformed 2,000-gene space, or gene-gene correlation agreement) so the flow's noise residual is actually tested.

[low] references

The C2ST real-vs-real reference is somewhat above 0.5 and moves with the seed's training subsample by about 0.02. This puts a floor of roughly 0.57-0.59 on the AUC and adds about 0.02 of noise. It is small next to the 0.25-0.35 gaps to the models, so the conclusions hold. The MMD references are small and n_compared = 540 is adequate.

Evidence: c2st_auc_reference 0.5855 (s0) vs 0.5664 (s1); mmd2_reference 0.00026 / -0.00028; n_compared 540 for all four runs.

Recommendation: no plan-level fix: state as a limitation (treat C2ST differences under about 0.02 as noise).

[medium] seed stability

The C2ST and MMD ranking (flow better than VAE) holds on both seeds: seed spread is about 0.02 against a model gap of about 0.07. But the models trade precision for recall, so no single 'best generator' follows. VAE precision also varies by 0.08 between seeds, which is as large as some gaps a reader might compare. With two seeds, every conclusion rests on n=2.

Evidence: precision/recall: vae_nb_s0 0.7907/0.8074, vae_nb_s1 0.8741/0.8000, flow_pca_s0 0.5611/0.987, flow_pca_s1 0.5315/0.9833. c2st: vae 0.9259/0.9045, flow 0.8572/0.8352.

Recommendation: Report the trade-off: the flow covers the data but about 45% of its samples fall outside real-cell neighborhoods, while the VAE is more precise but covers less. Do not name a best generator from C2ST alone.

[medium] training adequacy

The flow reached its best validation loss at epoch 13-14 of 400. Validation NLL then rose steeply while training NLL kept falling, which is classic fast overfitting (or a learning rate too high for 1,944 cells x 20 dims with 195k parameters). The flow's results come from a model checkpointed very early. The VAE looks adequate: best epochs 74/80, after the 40-epoch KL warmup, with validation flat from about epoch 50.

Critique and revision (cont.)

Evidence: runs/flow_pca_s0.json: best_epoch 13, val_loss 27.31 at epoch 13 -> 39.14 at epoch 38, train loss 25.41 -> 21.62. runs/flow_pca_s1.json: best_epoch 14, val_loss 27.16 -> 31.63. n_parameters 194880. runs/vae_nb_s0.json: best_epoch 74, val about 1602-1604 from epoch 50 on; s1: best_epoch 80.

Recommendation: In plan.json, lower the flow's lr (e.g. 3e-4), reduce hidden_dim (e.g. 64) and/or n_layers, or add weight decay, then rerun. Each run took 0.03 min, so this fits easily in the budget.

[medium] memorization / privacy

There is no sign of copying, but the numbers say 'off-distribution', not 'safe'. dcr_ratio is 1.42-1.59, and too_close_fraction is 0-0.007, far below the 0.05 expected for new real cells. Both fit the high C2ST: the samples sit farther from real data than real data does. The objective says 'for sharing', and no metric here establishes a privacy property. DCR is also computed only in the 14-PC space.

Evidence: dcr_ratio / too_close_fraction: vae_nb_s0 1.4436/0.0, vae_nb_s1 1.4175/0.0019, flow_pca_s0 1.5933/0.0, flow_pca_s1 1.5531/0.0074. objective.json raw_text: '...generate realistic synthetic cells for sharing'.

Recommendation: Any sharing recommendation must say that no copying was detected in a 14-PC space, that this is not a privacy guarantee, and that the cells are not yet realistic (C2ST about 0.84-0.93). Do not present the large DCR as a privacy benefit.

[medium] unjustified choices

Several reasons are contradicted by the results. The case for the VAE as the representation assumes it beats PCA, but PCA wins on trustworthiness. The flow's hidden_dim reason says 128 avoids overfitting, yet the flow overfits by epoch 13. The pca_components reason says 20 components 'keeps the main PBMC structure', but they explain only 24% of the variance. The lr reasons are generic and would read the same on any dataset.

Evidence: objective.json decision_reason: 'a latent-variable count model ... which a simpler method like PCA cannot do'. plan.json vae_nb reason: 'gives a 10-d latent embedding and a count generator in one model'. flow_pca hidden_dim: 'a larger network would overfit 2,160 cells in 20 dimensions' (best_epoch 13/14). pca_components: '20 components ... keeps the main PBMC structure' (input_pca_variance_explained 0.2396). lr: 'Adam at 1e-3 has no prior loss curve on this run suggesting oscillation' / 'Adam at 1e-3; no earlier loss curve on this dataset indicates instability.'

Recommendation: Revise the flow's lr, hidden_dim and pca_components reasons, and ideally the values, using the observed loss curves and variance explained. Restate the VAE's role so it no longer depends on beating PCA as a representation.

Revision made in response

Plan revision 2: Critique of revision 1: (1) flow overfit by epoch 13-14 -> flow lr 1e-3 to 3e-4, hidden_dim 128 to 64, n_layers 8 to 6, rerun both flow seeds; (2) reasons contradicted by results (flow hidden_dim/pca_components/lr, VAE role relative to PCA) -> rewritten from the observed loss curves, variance explained and trustworthiness; (3) generation criterion

Critique and revision (cont.)

failed (C2ST 0.84-0.93 vs reference 0.57-0.59) -> the flow retrain is the one generator change attempted, and the failure will be reported if it persists; (4) VAE latent below the PCA baseline -> reported as found, PCA named as the better neighbor-preserving representation; VAE unchanged and reused. Not plan-level, so stated as limitations: evaluation space is a 14-PC subspace close to the flow's input (fixed by the pipeline), the C2ST reference sits around 0.57-0.59 with about 0.02 seed noise, precision/recall trade-off across n=2 seeds, and memorization metrics show off-distribution samples, not privacy.

Limitations

- Models trained on PCA scores are evaluated in a PCA space of the same data, which may favor them on the shared realism metrics (see 'How to read this report').

- Fidelity is measured in a 14-component PCA space (0.2258 of training variance); differences outside it are not measured.

- The raw data were not QC-filtered, so low-quality cells and possible doublets were modeled and evaluated as normal cells.

- Fidelity and memorization were measured only in a 14-component PCA of the 2,000 selected genes. Gene-gene structure outside those components, and the flow's independent-noise residual, were not tested.

- The C2ST real-vs-real reference itself sits above 0.5 and differs between seeds (0.5855 and 0.5664), so C2ST differences of that size are noise.

- Two seeds per model give a stability check, not an estimate of variability.

- The critic's findings were addressed by one revision (flow retrained, reasons corrected). The generation failure persisted after it, and no further generator changes were made under the one-revision rule.

- Memorization checks are evidence about copying, not a privacy guarantee. No membership-inference test was run.

Reproducibility

Command: ./run.sh pbmc "Learn a low-dimensional representation and generate realistic synthetic cells for sharing" --report-number 1

Plan seed 0; plan revision 2; budget 20 min; pipeline time 0.71 min

Preprocessing hash e17cdd914d50a7cc

python 3.13.5, torch 2.14.0, numpy 2.5.3, scikit-learn 1.9.1

- vae_nb_s0: seed 0, config hash 9fc8ea4100ade8b1, 0.192 min on cpu

- vae_nb_s1: seed 1, config hash f9c475f41d843b6e, 0.202 min on cpu

- flow_pca_s0: seed 0, config hash 6c3ae49bc620d947, 0.044 min on cpu

- flow_pca_s1: seed 1, config hash 53de5617ecf45e7f, 0.043 min on cpu

Generated 2026-10-25 12:33. All intermediate files: outputs/pbmc/.